Curai

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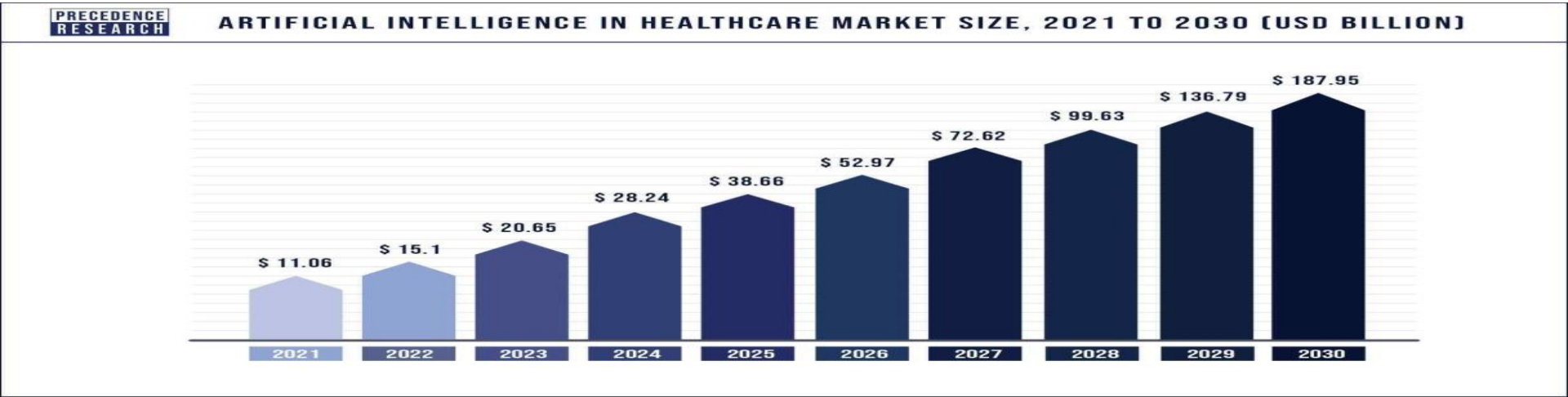
1. Introduction

Introduction

The healthcare sector plays a crucial role in improving quality of life and ensuring community well-being. However, despite having over 100,000 hospitals worldwide, we still face major challenges like overcrowding, staff shortages, and long waiting times.

But today, with technological advancements, we are witnessing a major transformation in healthcare, thanks to Artificial Intelligence 🤖 and data-driven solutions. These technologies enhance diagnostic accuracy, optimize resource management, and improve communication between doctors and patients.



Introduction

Why Did We Develop CurAi?

- Fast result
- smarter
- more efficient healthcare than ever before

2. Problem Definition

Problems that facing doctors:-

Overlapping Appointments:

1-Caused by manual or disorganized scheduling.

2-Leads to delays and disruption in doctors' schedules.

3-Takes away time and energy.

4-Difficulty communicating with patients about changes or reminders.

Problems facing Patients

1- Difficulty in identifying the right medical specialty based on their symptoms, which leads to delayed treatment or additional costs.

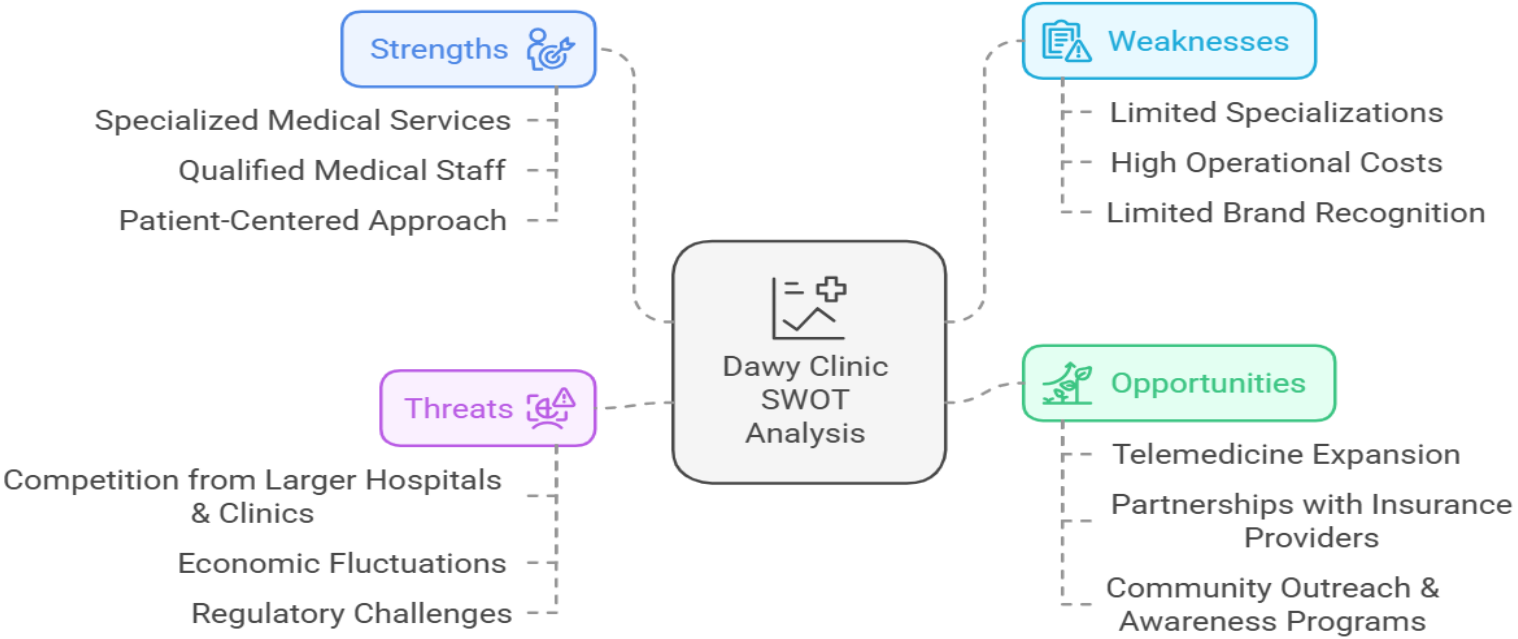
2- Challenges in booking appointments due to reliance on traditional methods, such as phone calls or visiting clinics in person.

3- Problems finding the nearest doctor during emergencies, which may worsen their condition.

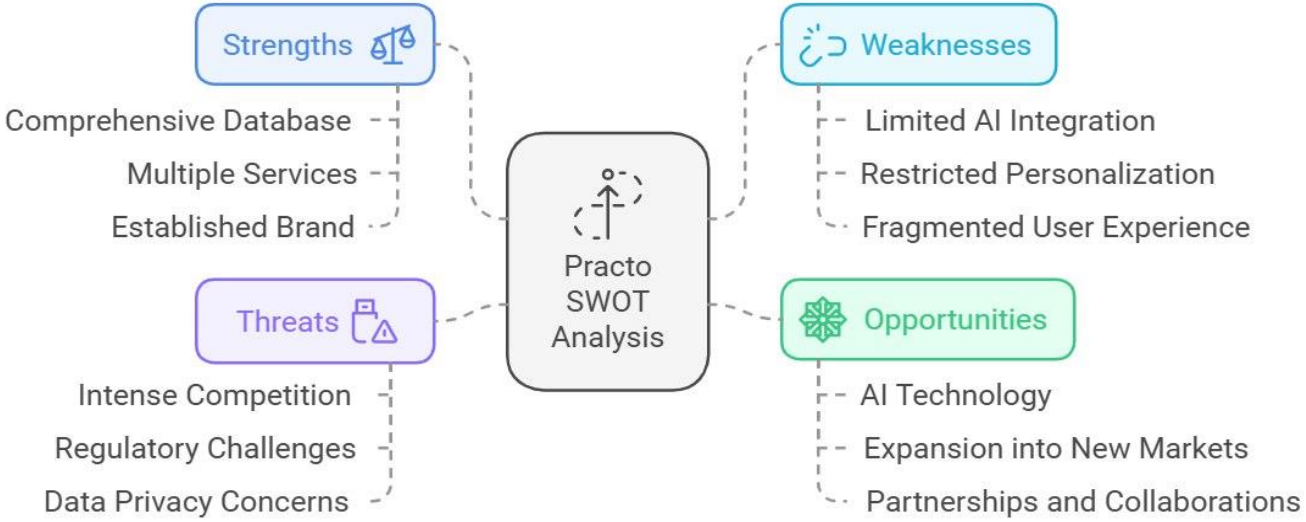
4- Additional expenses caused by booking errors or miscommunication.

3. Related Work

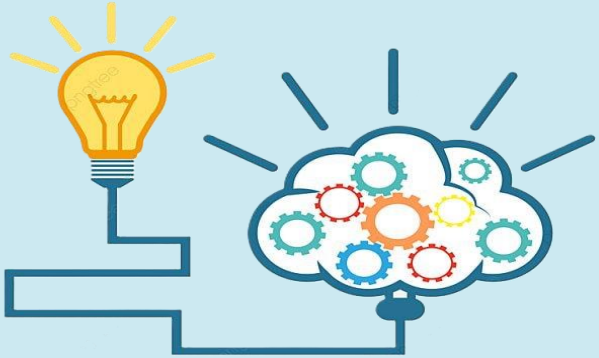
Related Work



Related Work



4. Proposed Solution



Proposed Solution

Responsive User-Friendly Design

Geographical

Chatbot

Cross-Platform Appointment Booking

Medical history

&

Bonus point

Real-Time Notifications

Solutions We Will Provide in Our Project

1- Responsive User-Friendly Design :

- Provides a simple, intuitive interface accessible across all devices.

2- Automated Scheduling System :

- Efficiently organizes appointments, reducing conflicts and manual errors.

3- Real-Time Updates:

- Doctors get instant alerts if a patient cancels or reschedules an appointment. This helps them plan their time better and avoid wasted slots.

Solutions We Will Provide in Our Project

4- Automated Notifications:

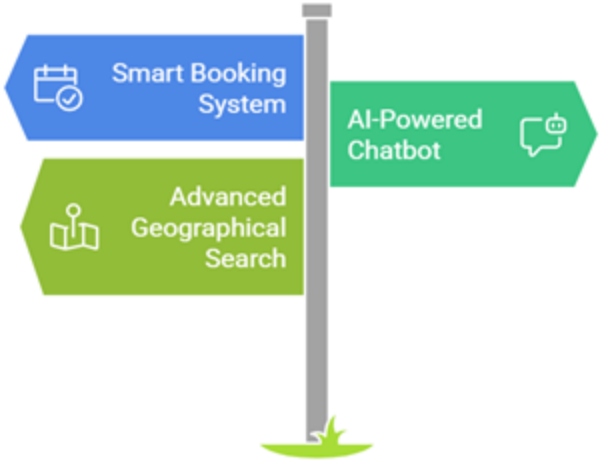
- Sends timely notifications to patients about appointment changes, ensuring clear and effective communication.

5- Advanced Geographical Search:

- Allows patients to find nearby doctors with ratings and cost comparisons.

6- AI-Powered Chatbot:

- Helps patients identify the right medical specialty based on their symptoms.



6. Business Model Canvas

Key Partnerships

- Cloud service providers.
- AI research institutions.
- Industry-specific partners for tailored solutions.

Key Activities

- AI model development and training.
- Marketing and customer acquisition.
- Continuous product improvement.

Key Resources

- AI algorithms and data infrastructure.
- Skilled development team.
- Cloud computing resources.

Value Propositions

- AI-powered personalized recommendations for users.
- Enhancing decision-making with data-driven insights.
- Providing tailored solutions for diverse user needs.

Customer Relationship

- Personalized customer support.
- Regular updates and feedback loops.
- Self-service tools and resources.

Channels

- Website and mobile app.
- Social media platforms.
- Partnerships with tech companies.

Customer Segments

- Patients
- Doctors
- Administrators
- Clinics & Hospitals

Cost Structure

- Cloud hosting and infrastructure costs.
- Development and research expenses.
- Marketing and customer support costs.

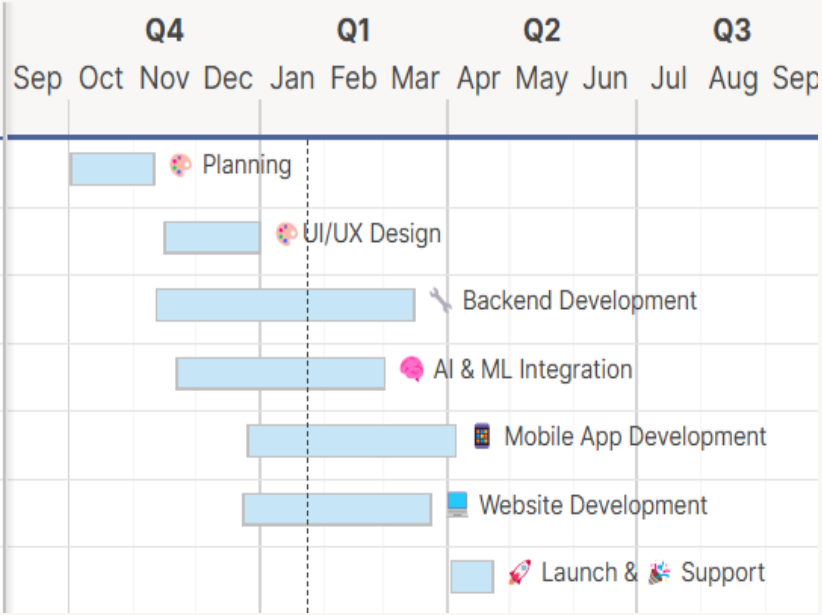
Revenue Stream

- Subscription model for premium features.
- One-time payment for specific services.
- Partnerships and collaborations.

7. Timeline Development

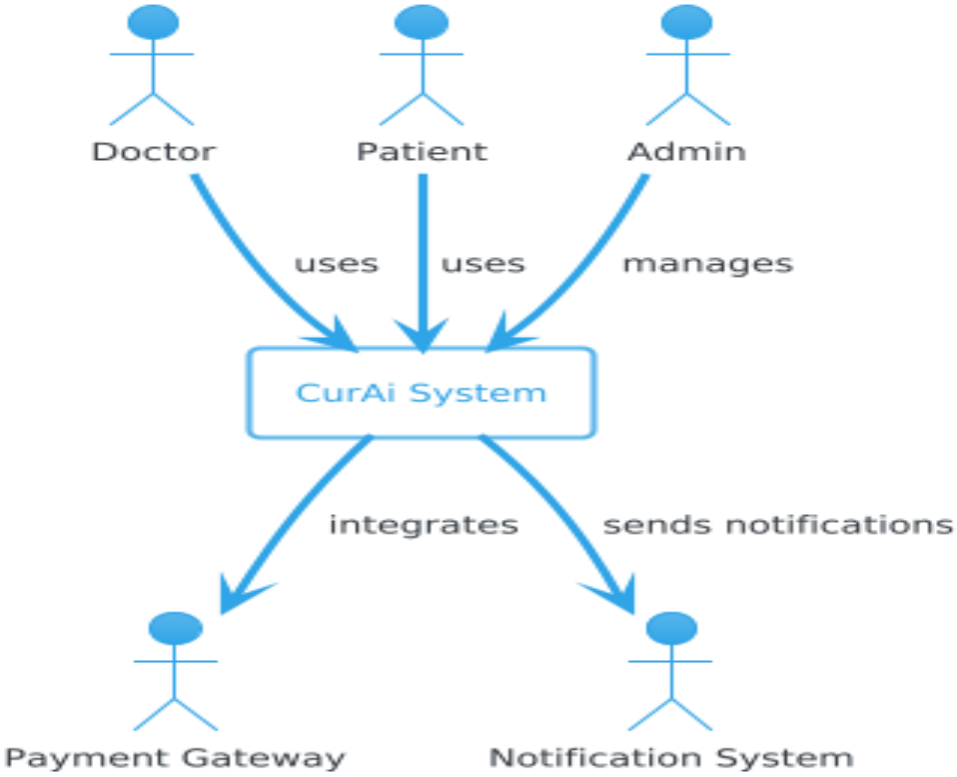
Timeline Development

Primary Column	Start Date	End Date
Planning	10/01/24	11/10/24
UI/UX Design	11/15/24	12/31/24
Backend Development	11/11/24	03/15/25
AI & ML Integration	11/21/24	03/01/25
Mobile App Development	12/25/24	04/04/25
Website Development	12/23/24	03/23/25
Launch & Support	04/02/25	04/22/25

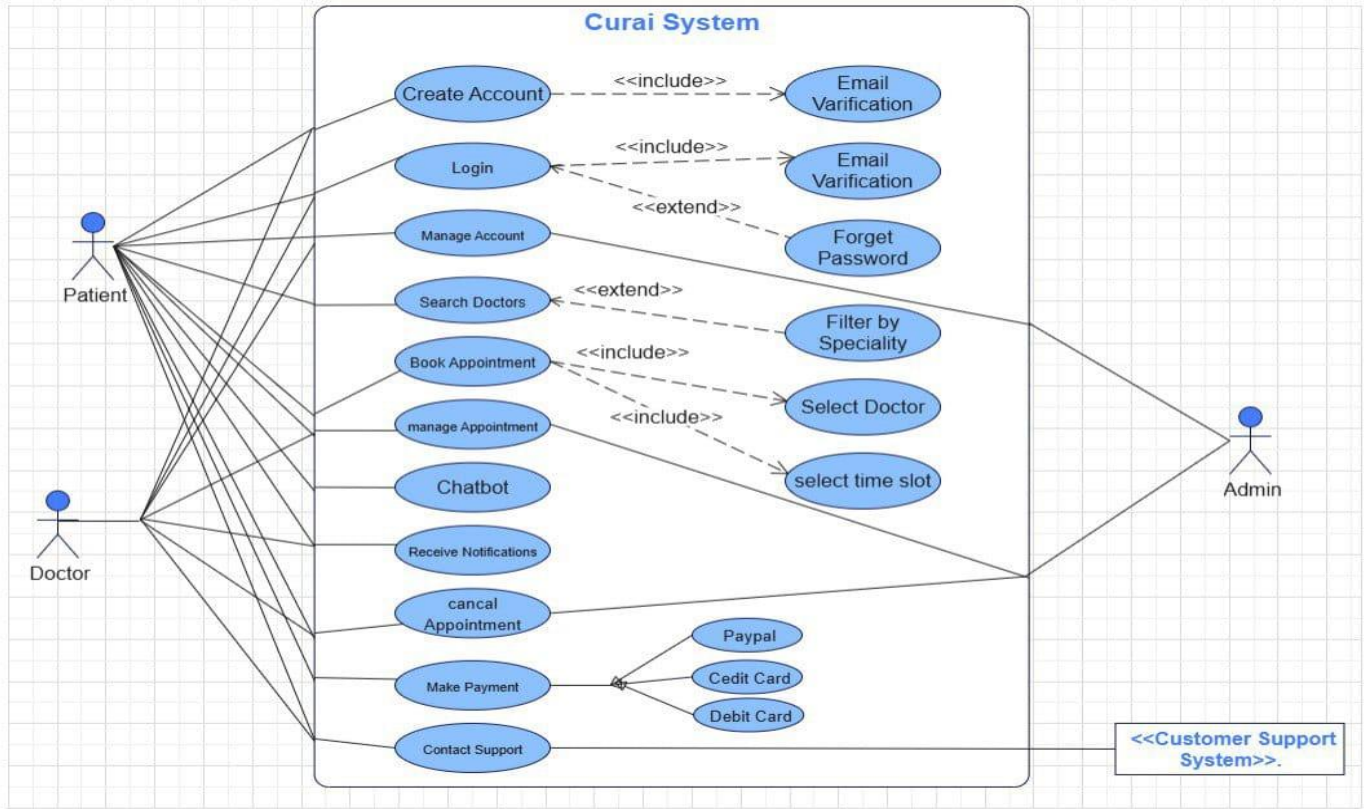


8. System Analysis

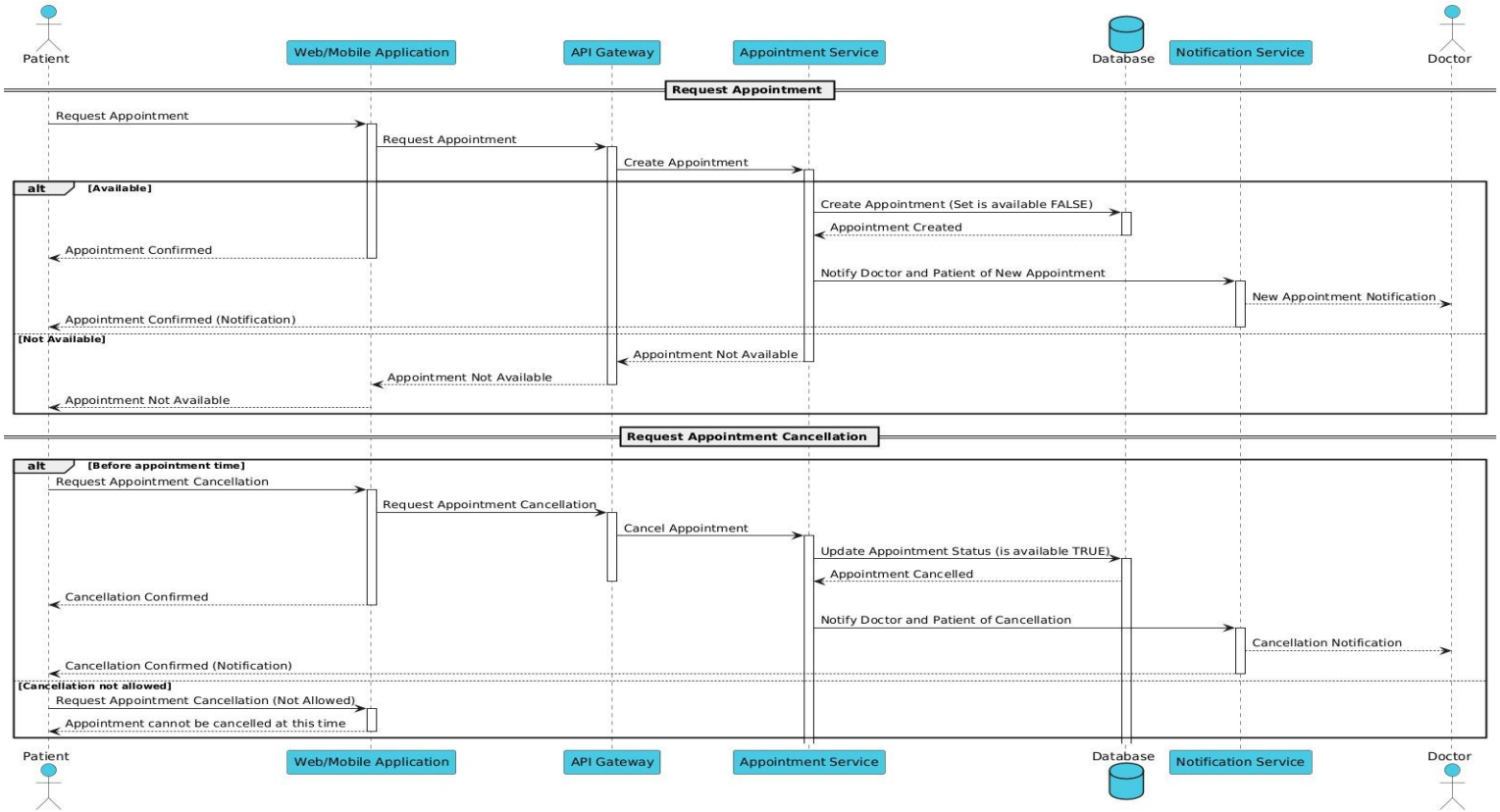
Context Diagram



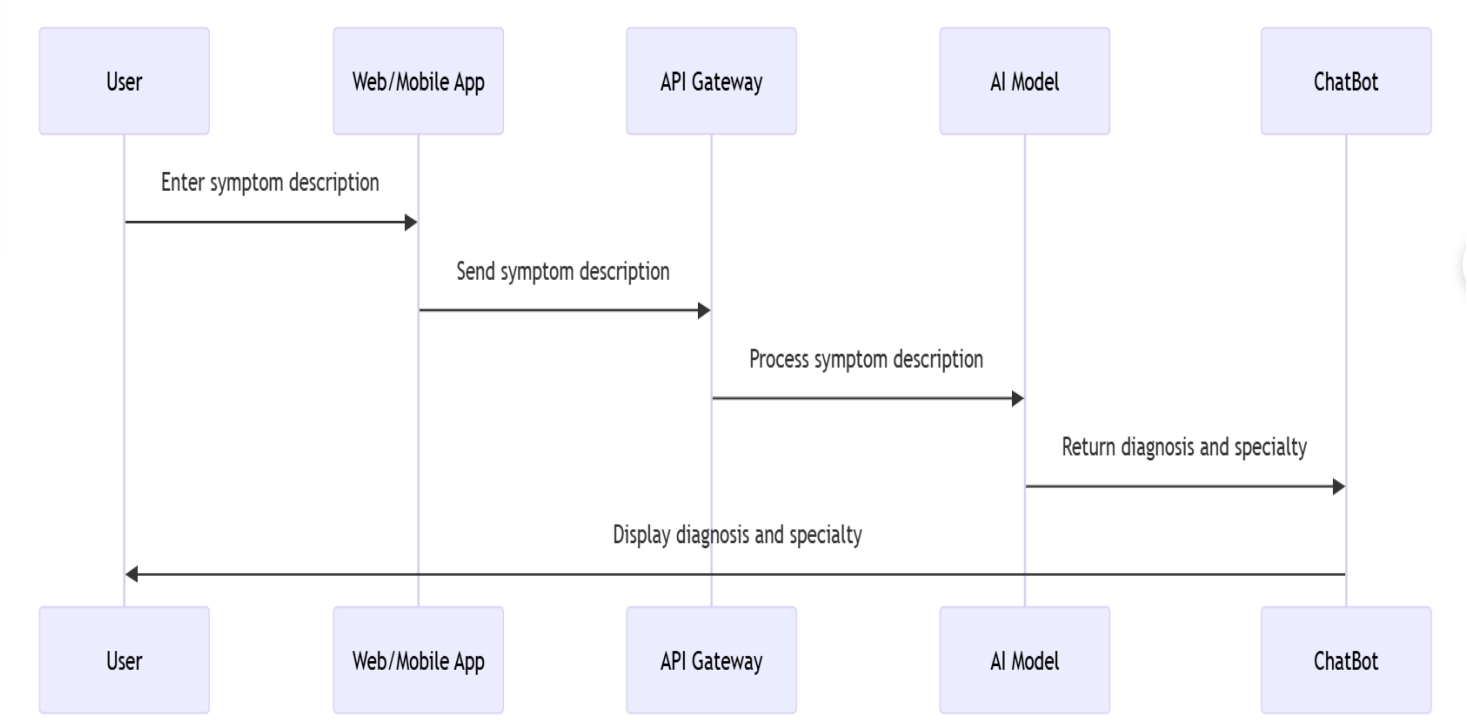
Use Case Diagram



Sequence Diagram for Booking & Cancling



Sequence Diagram for Chatbot



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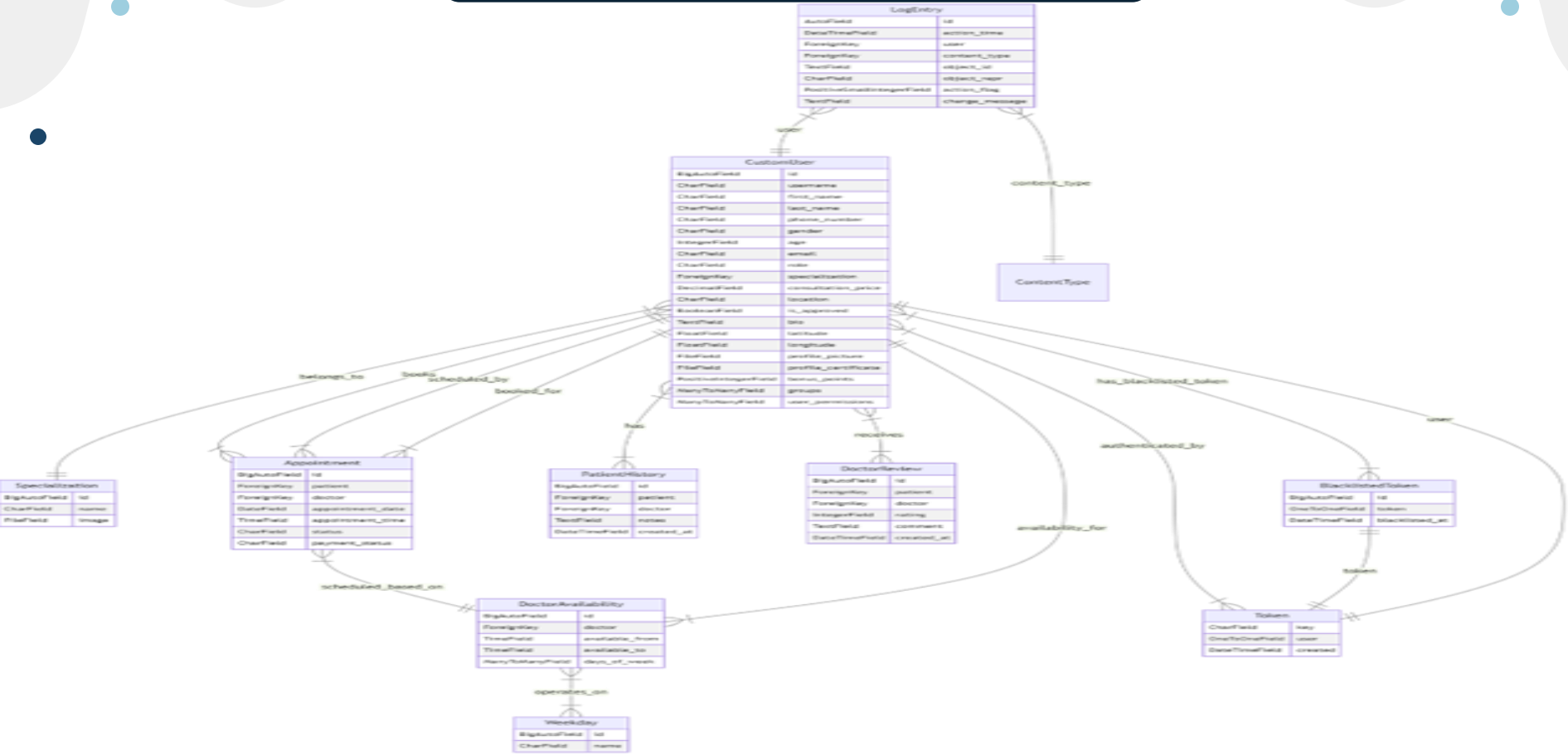
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DateTimeField action_time
ForeignKey user
ForeignKey content_type
TextField object_id
CharField object_repr
PositiveSmallIntegerField action_flag
TextField change_message
user
content_type
ContentType
CustomUser
BigAutoField id
CharField username
CharField first_name
CharField last_name
CharField phone_number
CharField gender
IntegerField age
CharField email
CharField role
ForeignKey specialization
DecimalField consultation_price
CharField location
BooleanField is_approved
TextField bio
FloatField latitude
FloatField longitude
FileField profile_picture
FileField profile_certificate
PositiveIntegerField bonus_points
ManyToManyField groups
ManyToManyField user_permissions
belongs_to
books
scheduled_by
booked_for
has
makes
has_blacklisted_token
authenticated_by
availability_for
user
Specialization
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CharField name
FileField image
Appointment
BigAutoField id
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ForeignKey doctor
DateField appointment_date
TimeField appointment_time
CharField status
CharField payment_status
PatientHistory
BigAutoField id
ForeignKey patient
ForeignKey doctor
TextField notes
DateTimeField created_at
DoctorReview
BigAutoField id
ForeignKey patient
ForeignKey doctor
IntegerField rating
TextField comment
DateTimeField created_at
BlacklistedToken
BigAutoField id
OneToOneField token
DateTimeField blacklisted_at
scheduled_based_on
DoctorAvailability
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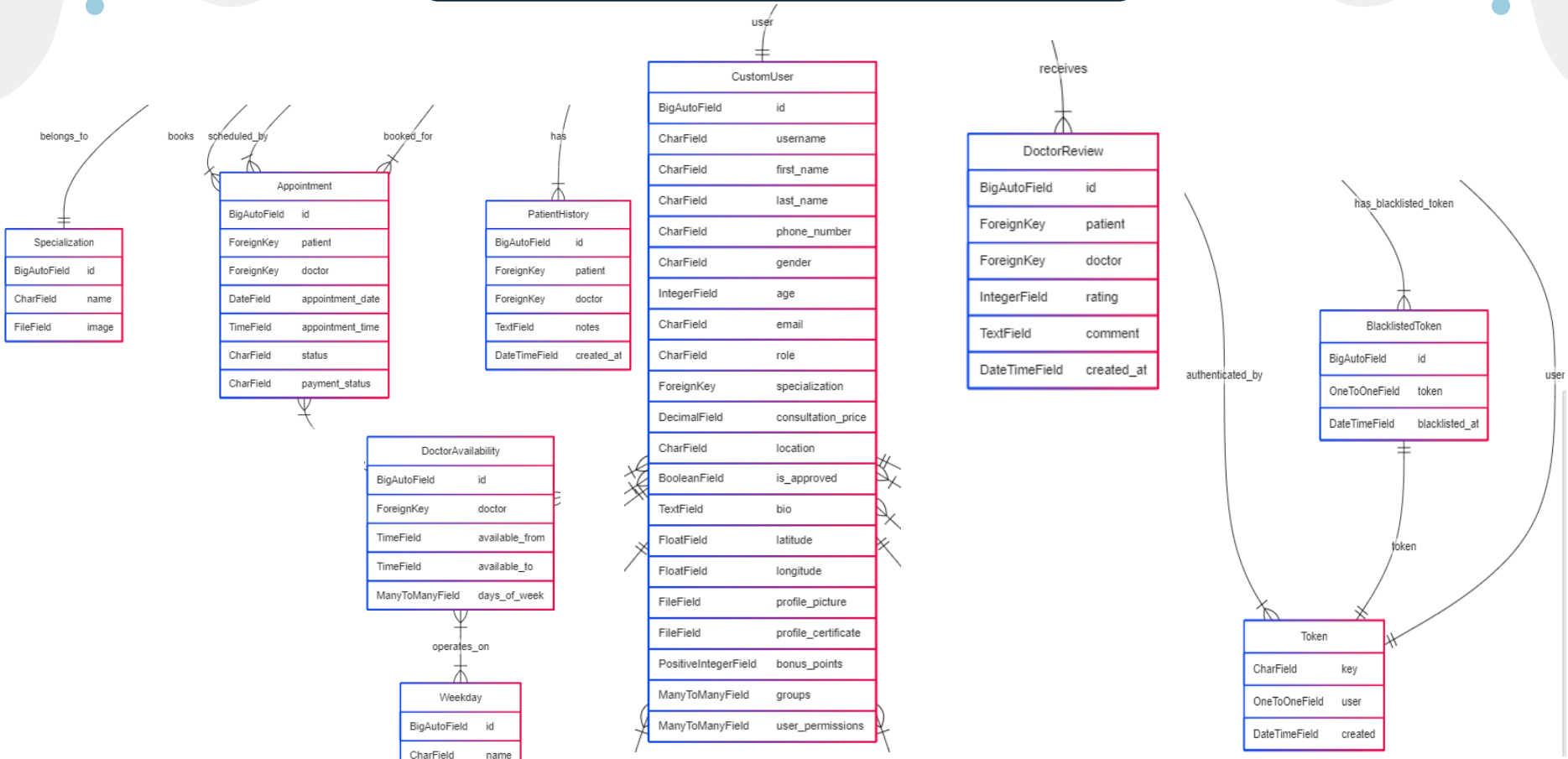
ForeignKey doctor
TimeField available_from
TimeField available_to
ManyToManyField days_of_week
operates_on
Weekday
BigAutoField id
CharField name
token
Token
CharField key
OneToOneField user
DateTimeField created

ERD


user
receives
belongs_to
books
scheduled_by
booked_for
has
has_blacklisted_token
authenticated_by
user
token
operates_on
CustomUser
BigAutoField id
CharField username
CharField first_name
CharField last_name
CharField phone_number
CharField gender
IntegerField age
CharField email
CharField role
ForeignKey specialization
DecimalField consultation_price
CharField location
BooleanField is_approved
TextField bio
FloatField latitude
FloatField longitude
FileField profile_picture
FileField profile_certificate
PositiveIntegerField bonus_points
ManyToManyField groups
ManyToManyField user_permissions
DoctorReview
BigAutoField id
ForeignKey patient
ForeignKey doctor
IntegerField rating
TextField comment
DateTimeField created_at
Specialization
BigAutoField id
CharField name
FileField image
Appointment
BigAutoField id
ForeignKey patient
ForeignKey doctor
DateField appointment_date
TimeField appointment_time
CharField status
CharField payment_status
PatientHistory
BigAutoField id
ForeignKey patient
ForeignKey doctor
TextField notes
DateTimeField created_at
DoctorAvailability
BigAutoField id
ForeignKey doctor
TimeField available_from
TimeField available_to
ManyToManyField days_of_week
Weekday
BigAutoField id
CharField name
BlacklistedToken
BigAutoField id
OneToOneField token
DateTimeField blacklisted_at
Token
CharField key
OneToOneField user
DateTimeField created

5. Implementation

Ai Technologies

Programming Language:

- **Python:** Core language for AI development

Libraries & Frameworks:

- **scikit-learn:** Model training (SVM, Random Forest, etc.)
- **Nltk:** NLP preprocessing (tokenization, lemmatization, etc.)
- **TF-IDF Vectorizer:** Convert text to numeric features
- **pandas & NumPy:** Data cleaning and manipulation

Backend Technologies

Framework:

- Django

Task Queue:

- Celery

Authentication:

- JWT (JSON Web Tokens)

API:

- Django REST Framework (DRF)

Database:

- SQLite (Development)
- PostgreSQL (Production)

- Password validation and secure hashing methods (e.g., Argon2, PBKDF2).

Web Technologies

Technologies & Architecture

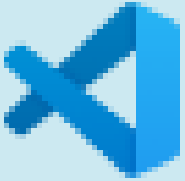
- Vite – Fast build tool for development
- React JS – For building UI components.
- Tailwind CSS – Utility-first CSS framework.
- React Router DOM – Client-side routing.
- Formik – Form handling.
- Yup – Form validation.
- Component-Centric Architecture – Each component has its own folder with logic and UI in one file.
- Axios – For making HTTP requests

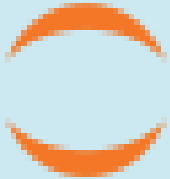
Mobile Technologies

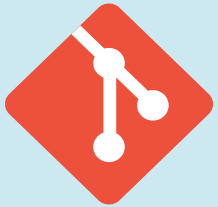
Technologies & Architecture

- Dart – as programming language
- Flutter - as framework to build UI
- BloC(Cubit) – as state management
- Clean Architecture – as architecture pattern
- Dio – as networking
- OSMap – as pick location Doctor
- Paymob – as payment
- Sentry – as Error Monitoring

Development Tools







5.Dataset

Dataset Sources and Their Roles

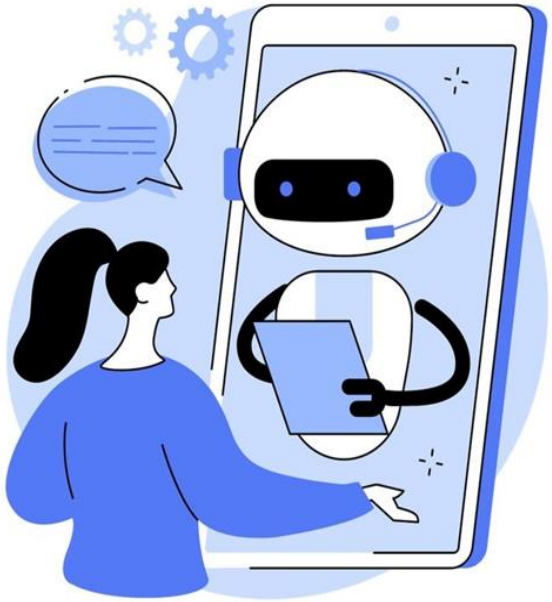
We used two datasets to train our AI chatbot:

Medical Dataset

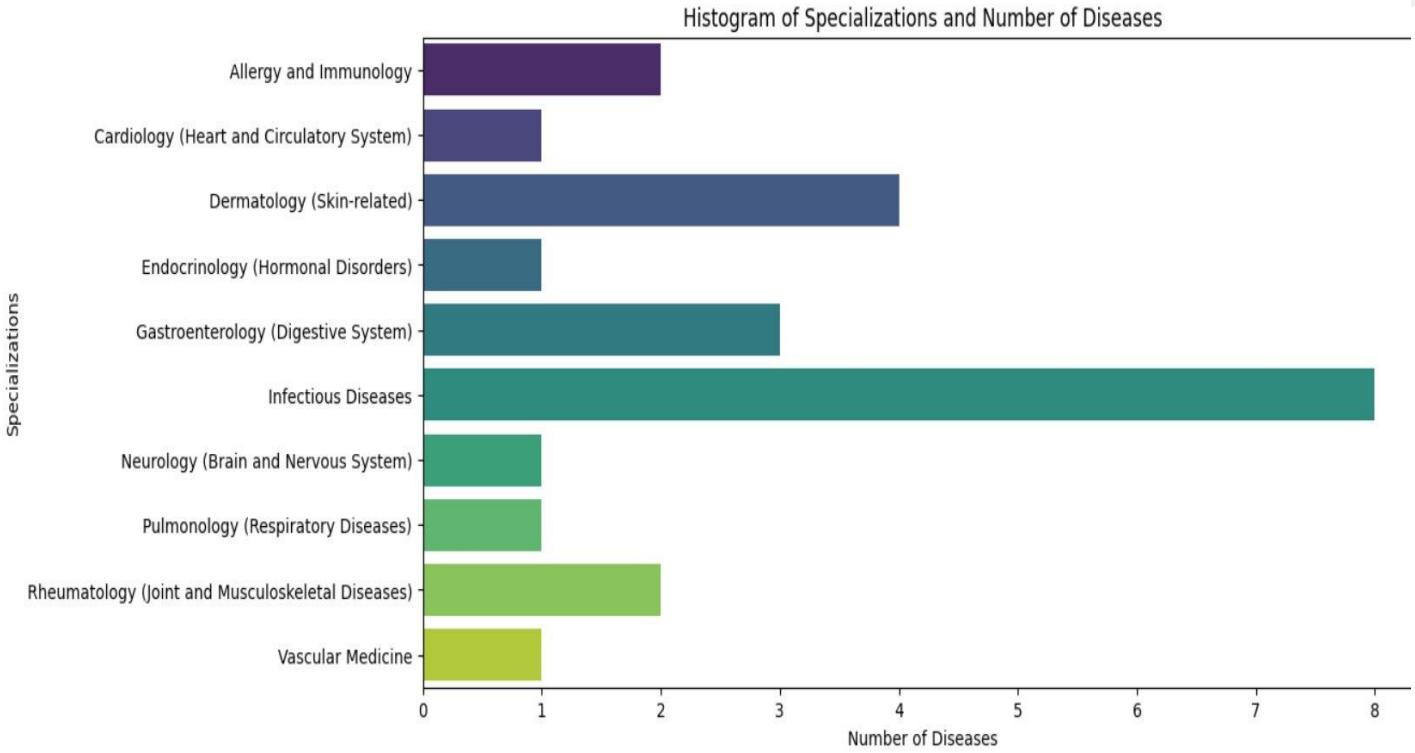
- Contains symptoms and disease names
- Specialty added manually
- **Used for disease prediction**
- Source: **Kaggle** – https://www.kaggle.com/datasets/niyarrbarman/symptom2disease/data

Non-Medical Dataset

- General text (sports, engineering, etc.)
- **Used to detect and filter non-medical inputs**
- Source: **Kaggle** – https://www.kaggle.com/datasets/rmisra/news- category-data

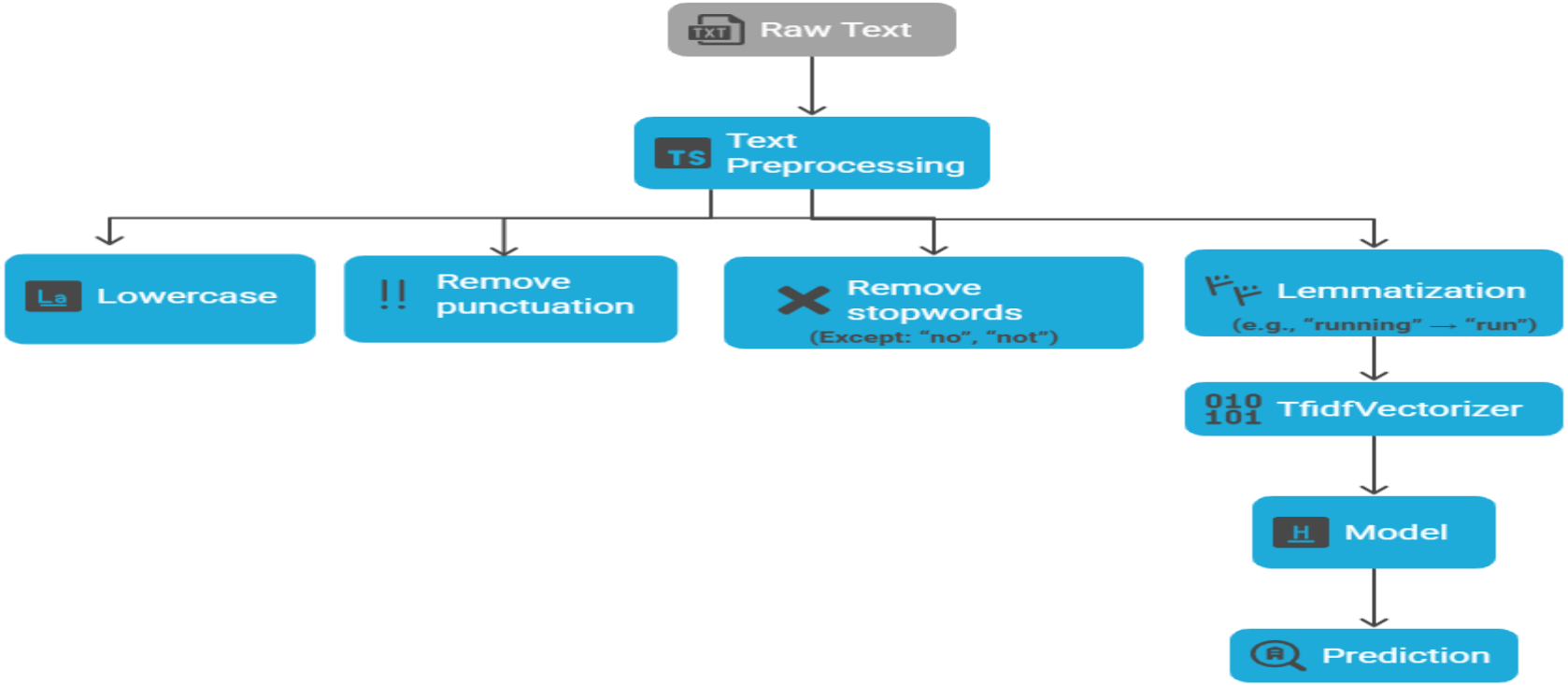


Number of Diseases per Medical Specialty



This chart displays the number of unique diseases categorized under each of the 10 medical specialties in the dataset.(The dataset contains 24 diseases distributed across 10 medical specialties)

Steps of Text Cleaning & Preparation

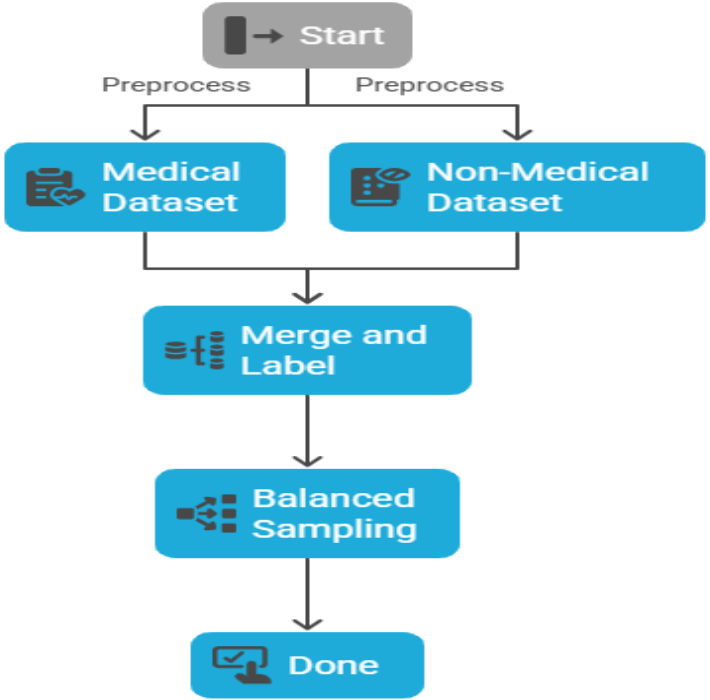


This diagram shows our full text classification workflow:
From cleaning and tokenizing text, applying TF-IDF for vectorization.

Data Preparation and Integration

- We used a medical dataset that contains symptoms and disease names, and we manually added the medical specialty.

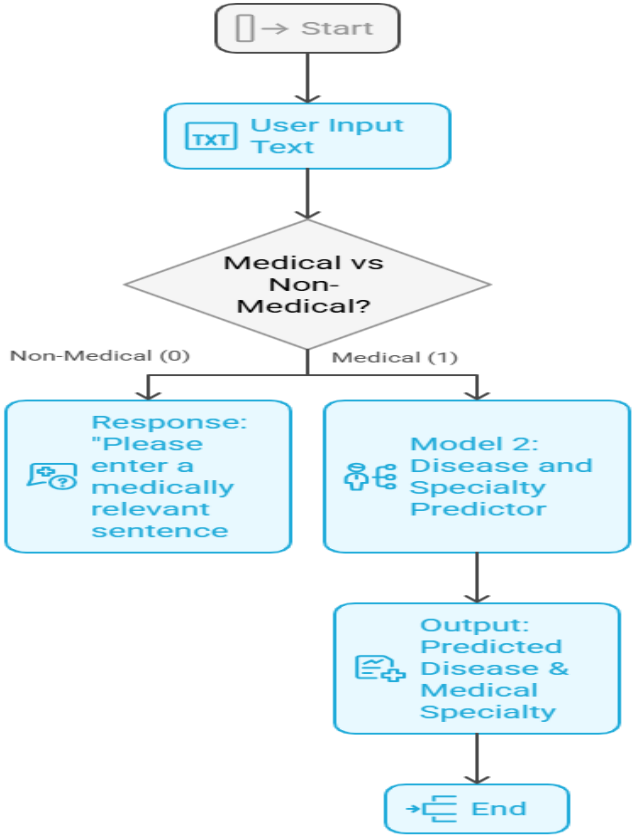
- We merged it with a non-medical dataset from various domains, while ensuring balanced sampling to avoid bias or overfitting.



Medical Text Filter

Multinomial Naive Bayes classifier

- chosen for its high efficiency in medical text classification, effectively handling categorical features like symptom keywords and medical terms.
- Achieved **99%** accuracy in classifying medical vs. non-medical text
- We used Supervised Learning for the first model, where the input text was labeled as:
- **1 for Medical text , 0 for Non-medical text**



Best Performing Model

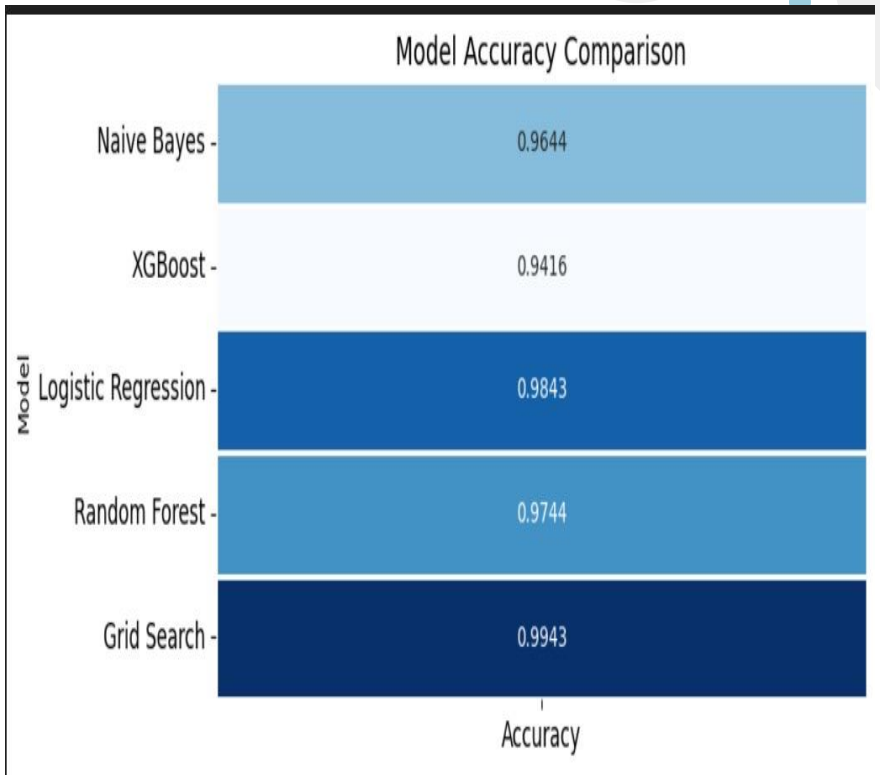
Several models were tested using **Grid SearchCV**.

SVM gave the best results with:

- Train Accuracy: **98%**
- Test Accuracy: **99%**

We also used **Supervised Learning** for disease prediction.
Medical inputs were labeled with:

1. **Disease Name**
2. **Medical Specialty**

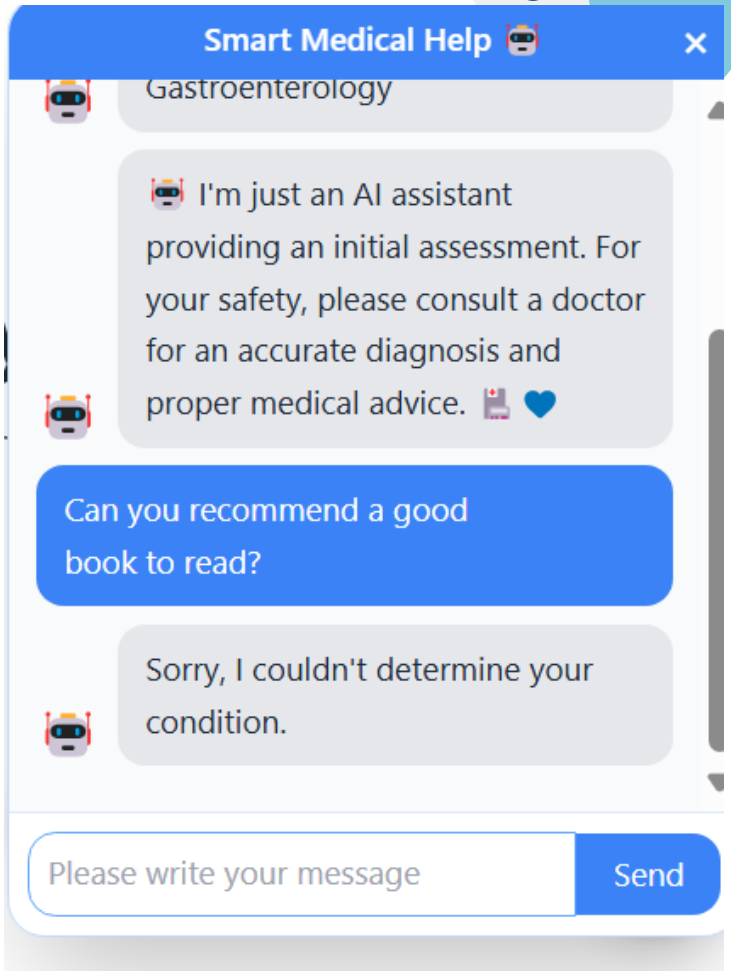


Model Comparison and Why We Chose SVM

Characteristic	SVM
Accuracy	Achieved top performance: Validation: 0.9918 – Test: 0.9943
Text Length	Handles short symptom inputs effectively due to strong generalization
Dataset Size	Unlike deep models, SVM works well with small, labeled datasets.
Performance	Surpassed Logistic Regression, Random Forest, and XGBoost in stability and accuracy
Complexity	Easy to train, tune, and understand with a linear kernel

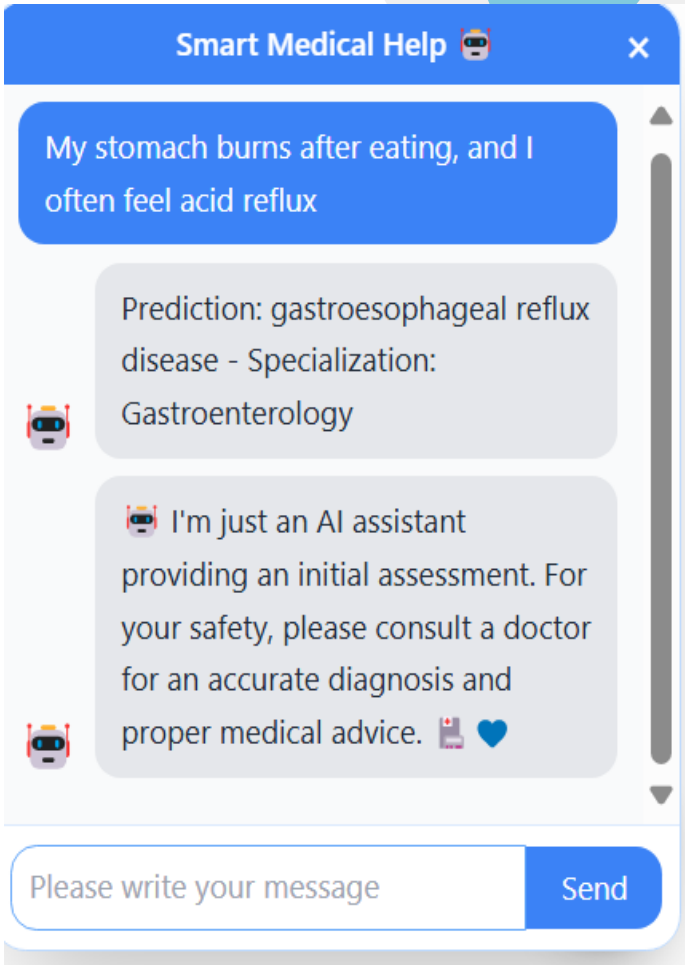
Chatbot Response – Non-Medical

The chatbot analyzes user input using a trained model to check medical relevance. If unrelated, it politely prompts the user to enter health-related symptoms.



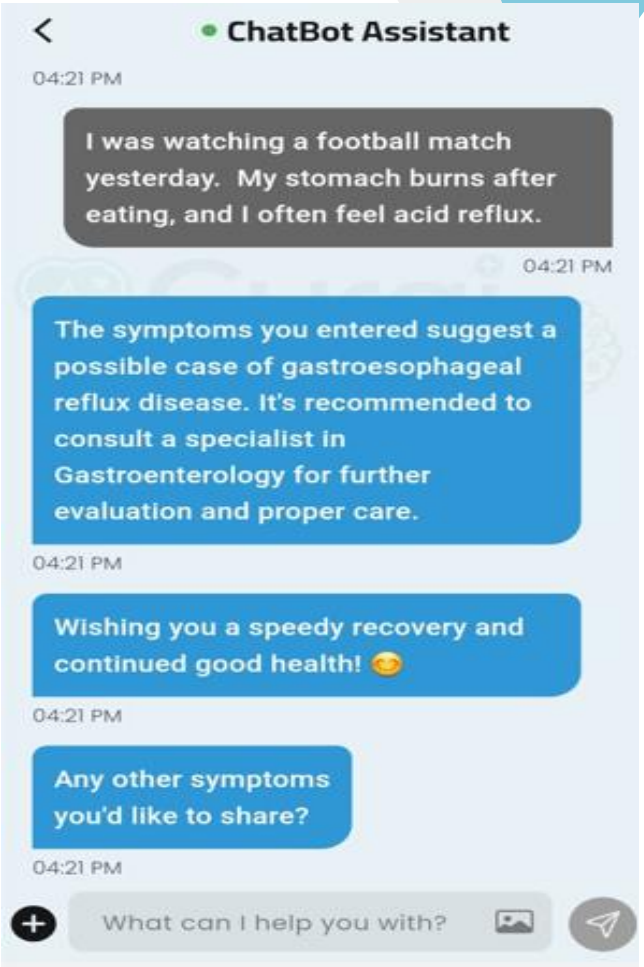
Chatbot Response – Medical Input

When symptoms are detected, the AI identifies the probable condition and directs the user to the relevant medical specialty for appropriate guidance.



Chatbot Response – Mixed Input

To test performance, a mixed-input sentence was entered. The chatbot accurately detected medical terms, ignored unrelated parts, and correctly identified the disease name.



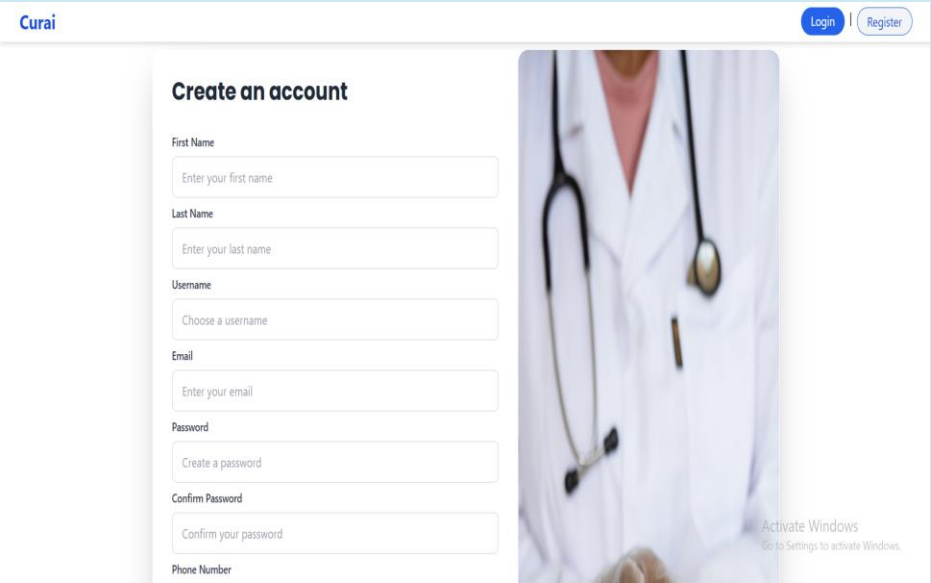
11. demo

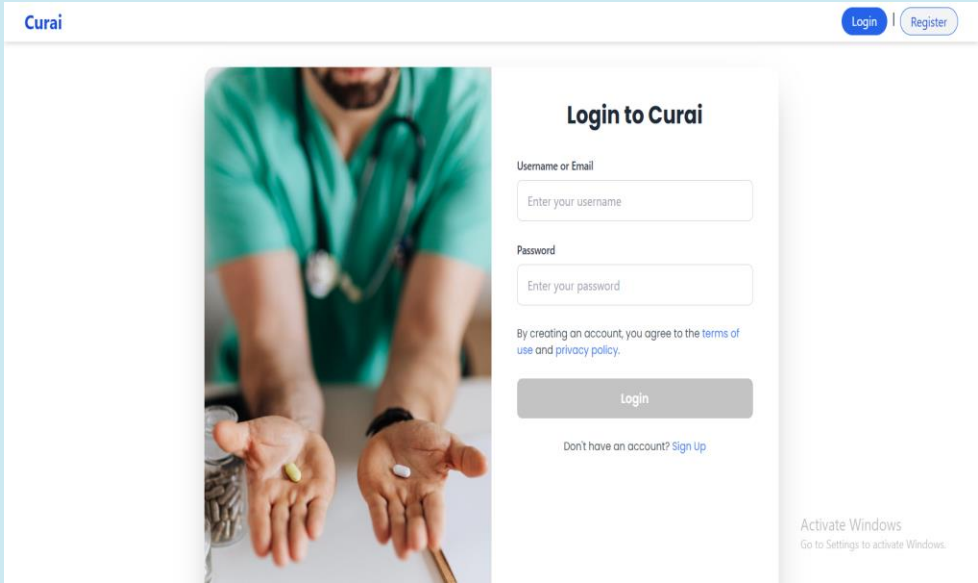
Web App

Create Account & Login

The application features separate pages for **"Create Account"** and **"Login"**.

- During the login process, a **token** is generated and stored in the browser's internal memory.
- If the token exists, the user is automatically redirected to the **home page**.
- If no token is found, the user is redirected to the **Login page** to **authenticate**.



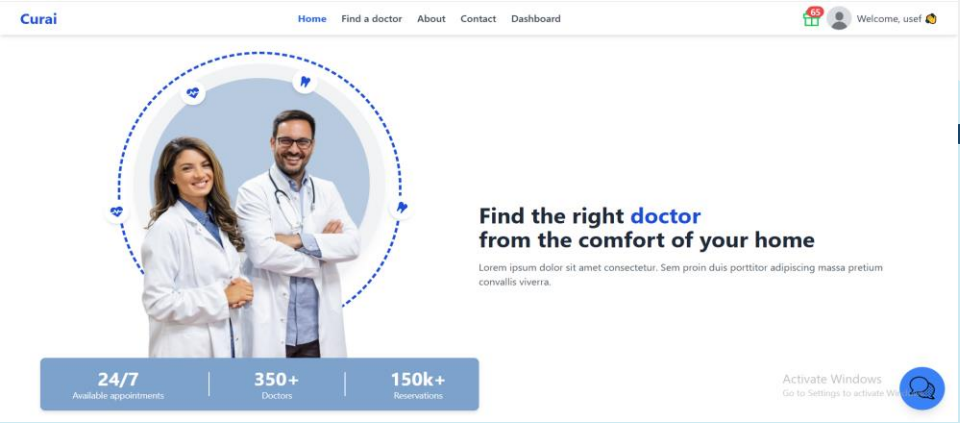


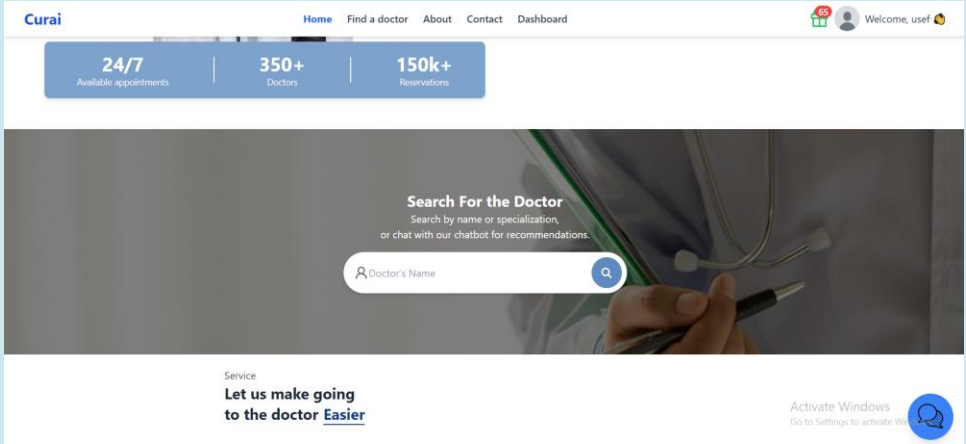
Home Page

1. The **Home page** serves as the main landing page after a successful login and includes the following sections:

2. **Hero Section**:
   A large image or banner displayed at the top of the page, visually introducing the platform and its purpose.

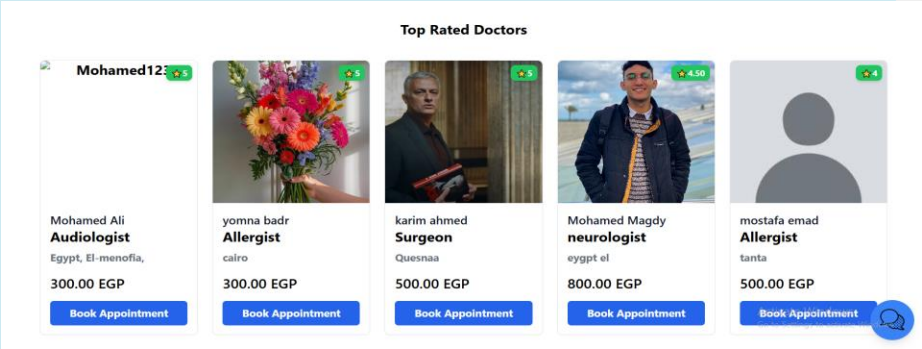
3. **Search Section**:
   A search input that allows users to look for specific doctors by name .

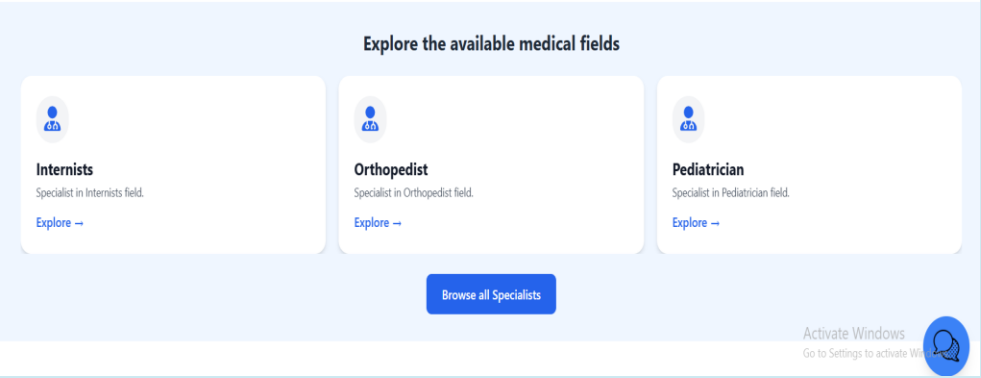


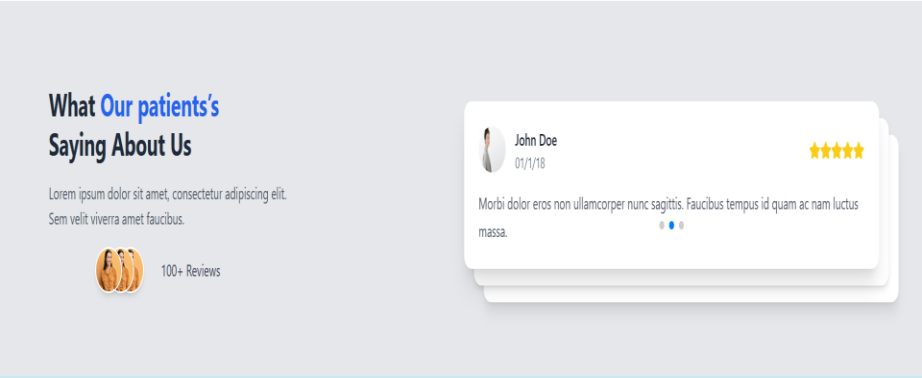


Home page

1. **Top Rated Doctors Section**:
   A curated list showcasing the highest-rated doctors on the platform based on user feedback and reviews.
2. **Explore Medical Fields Section**:
   A section displaying available medical specialties, helping users easily navigate and select a field relevant to their needs.
3. **Feedback Section**:
   Displays user feedback and testimonials about their experience with the platform and the doctors they consulted.







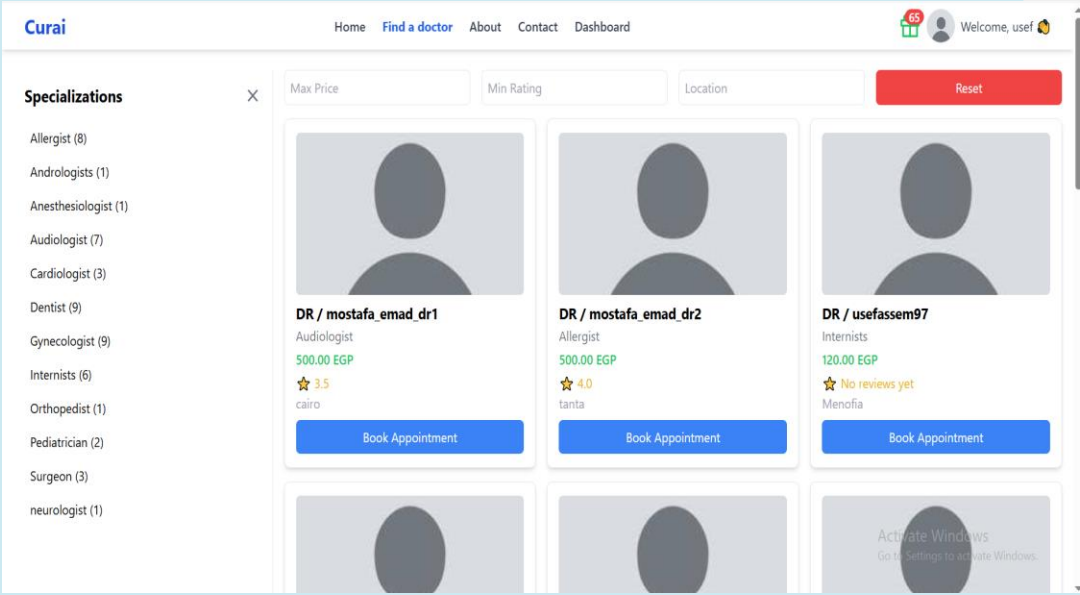
Doctors Page

- The **Doctors page** displays a list of all available doctors on the platform.
  This page includes the following features:

- **Doctors List :**
  A dynamic listing of registered and approved doctors, each displayed with key details such as name, specialty, consultation price, and rating.

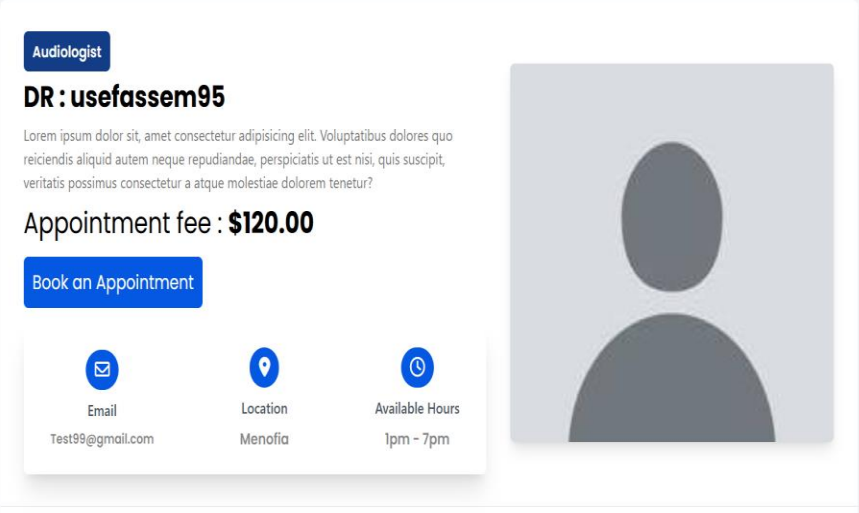
- **Filtering Options :**
  - **Specialization**
  - **Consultation Price**
  - **Rating**
  - **Location**

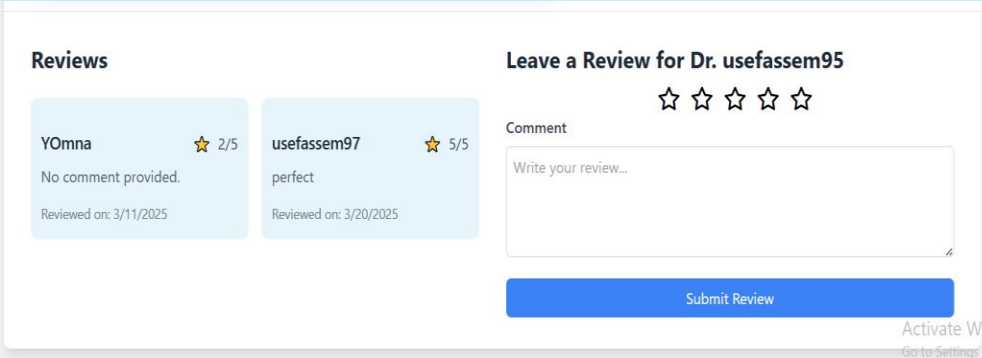
This allows users to efficiently find doctors
that match their preferences and needs.



Doctor Details

1. **Doctor Details Page**
2. Displays detailed information about the selected doctor.
3. Includes a **"Book an Appointment"** button for easy scheduling.
4. Shows the doctor's **email**, **location**, and **available hours**.
5. Contains a **reviews section** with patient feedback, including star ratings, comments, and review dates.
6. Provides a **review form** where users can rate and comment on the doctor's service.
7. Enhances trust by allowing new patients to read others' experiences before booking.





Booking system

1. **Booking System**

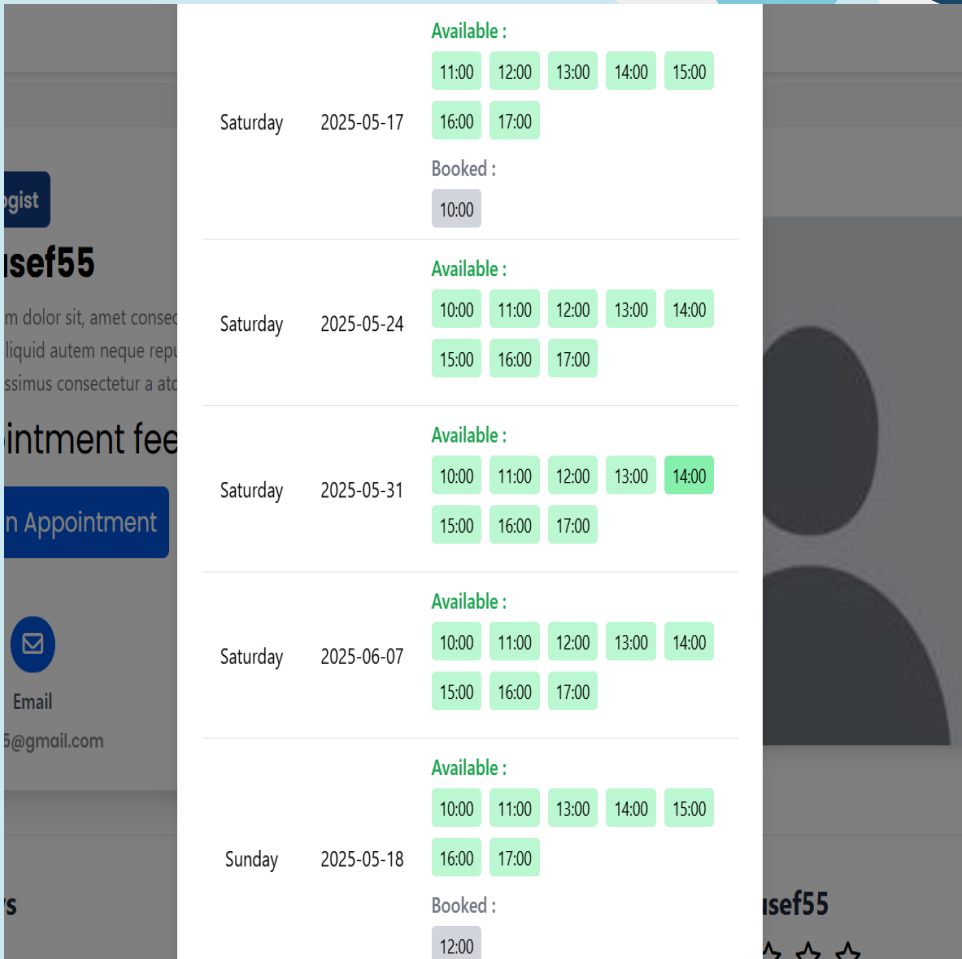
2. When the user clicks on "Book an Appointment", a **modal** appears displaying the available and booked time slots for the selected doctor.

3. **Booked slots** are visible but **disabled**, ensuring users cannot select them.

4. Only the **available time slots** are clickable, allowing users to choose a suitable appointment time.

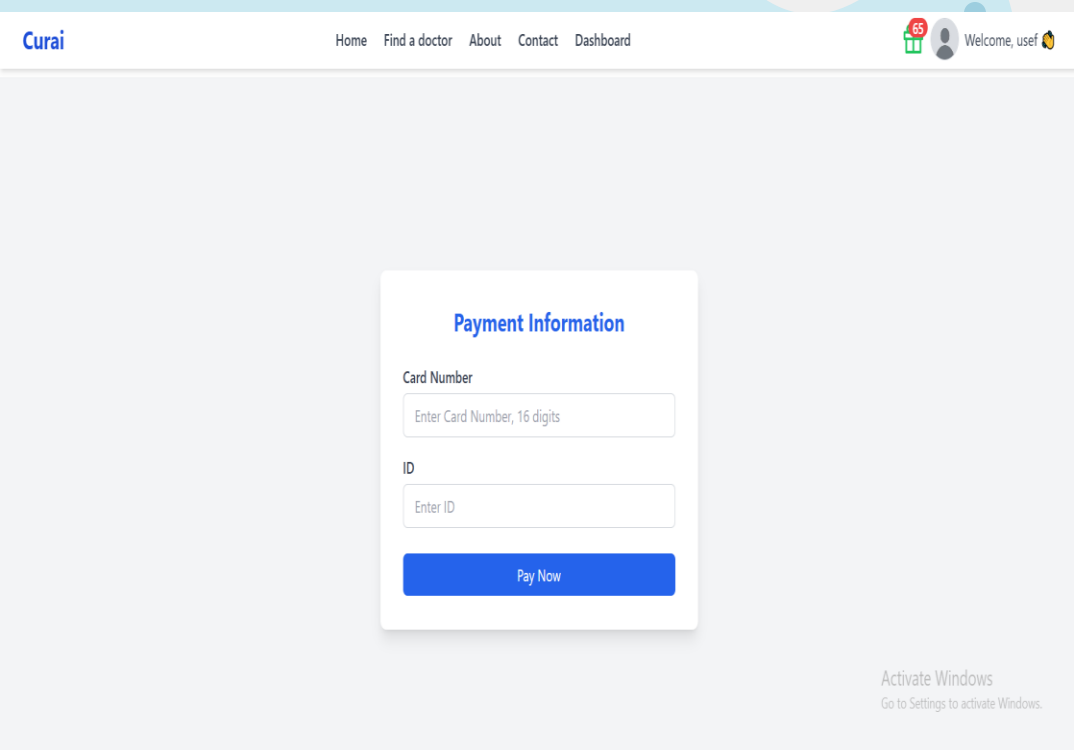
5. After successfully booking an appointment, the system generates a **unique booking ID** as confirmation.

6. This **booking ID** is used later during the **payment process** to finalize the reservation.



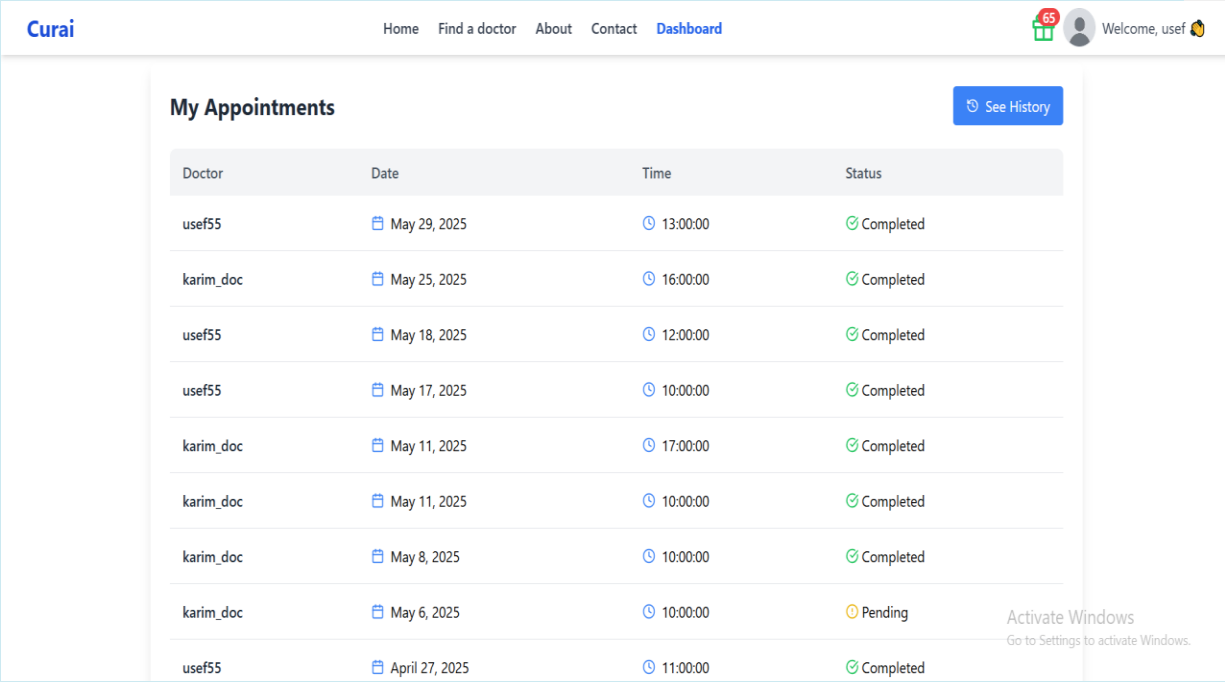
Payment page

1. **Payment Page**
2. This page is used to **confirm the appointment** by completing the payment.
3. The user enters the **Booking ID** that was generated during the appointment reservation process.
4. Once the payment is successfully completed, the **appointment is confirmed**.
5. The confirmed appointment is then **displayed on the Patient's dashboard** for future reference.
6. Additionally, the user's **bonus balance increases** as a reward for booking.



Patient Dashboard

1. **Patient Dashboard**
2. Displays a list of the patient's **upcoming appointments**.
3. For each appointment, it shows the **doctor's name** and appointment details.
4. Includes a **"See History"** button that allows the patient to view their **medical history**.
5. The medical history section contains all the **prescriptions written by doctors** during previous consultations.



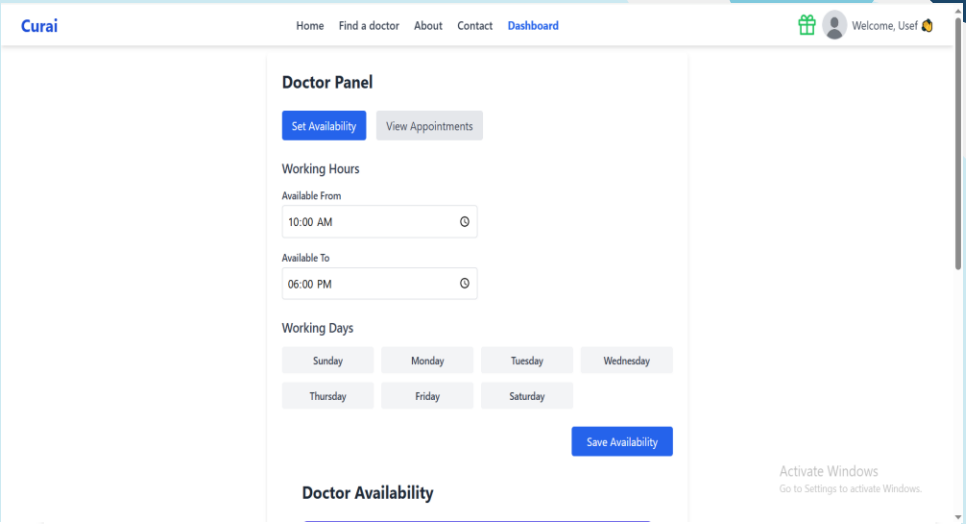
Doctor Dashboard

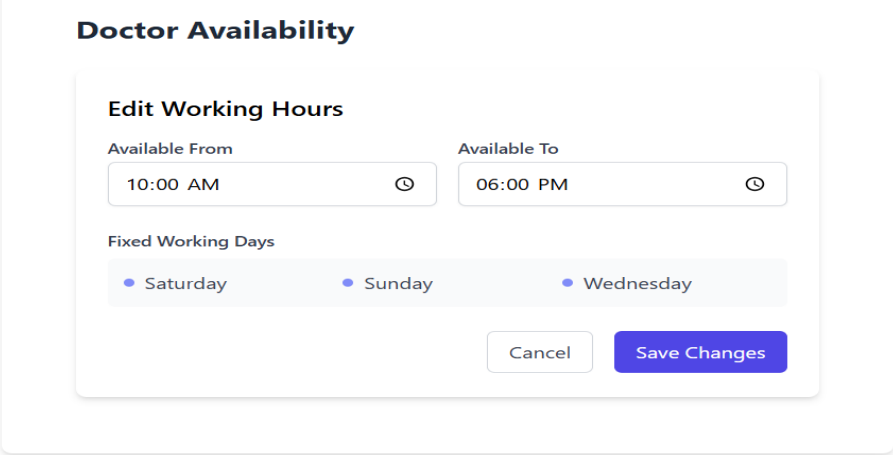
1. **Doctor Dashboard:**

2. **Schedule Management:** Doctors can set and manage their availability by adding preferred working hours.

3. **Edit Availability:** The dashboard allows doctors to modify or delete their working hours at any time, providing full flexibility to adjust or remove appointments as needed.

4. **User-Friendly Interface:** Designed for ease of use, making it simple for doctors to update, modify, or delete their appointment slots.



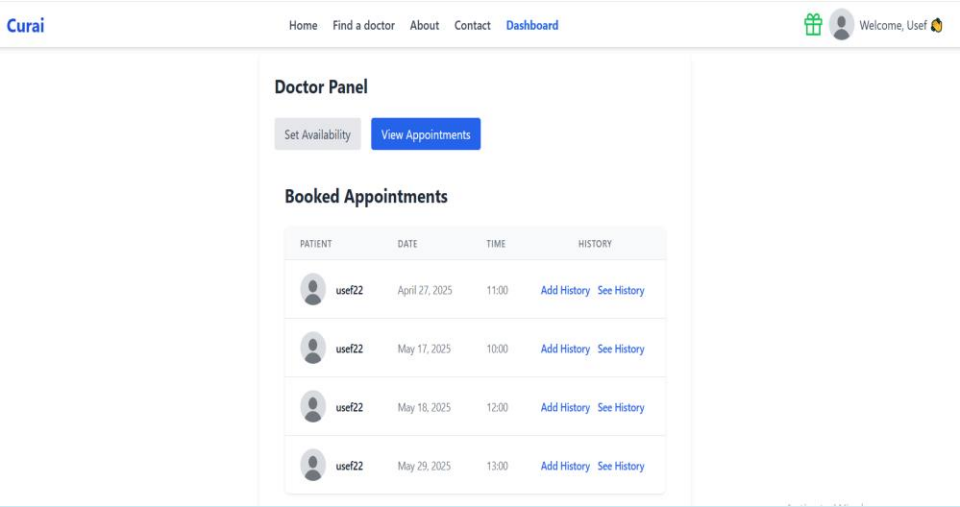


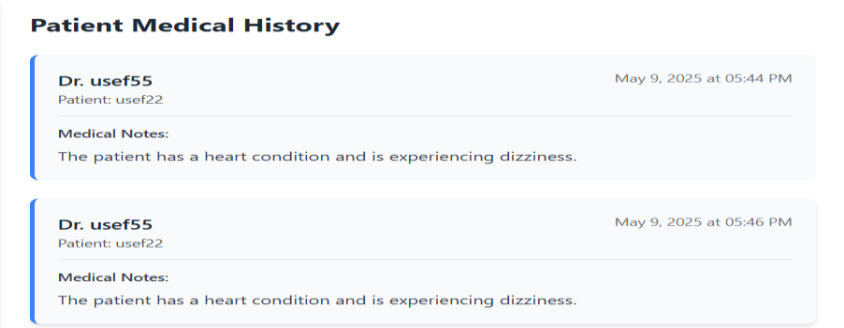
Doctor Dashboard

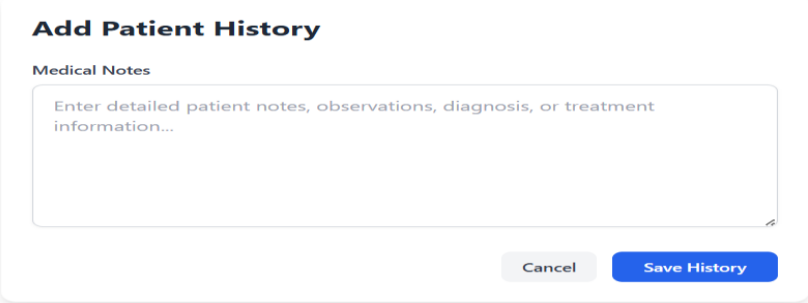
1. **View Booked Appointments:**
   Doctors can see a list of all appointments booked with them, including appointment times and patient names.

2. **Access Patient History:**
   For each booked appointment, doctors can view the patient's medical history to better understand their background before the consultation.

3. **Add Diagnosis Results:**
   After finishing a consultation, doctors can update the patient's history by adding the diagnosis, notes, or treatment plan.







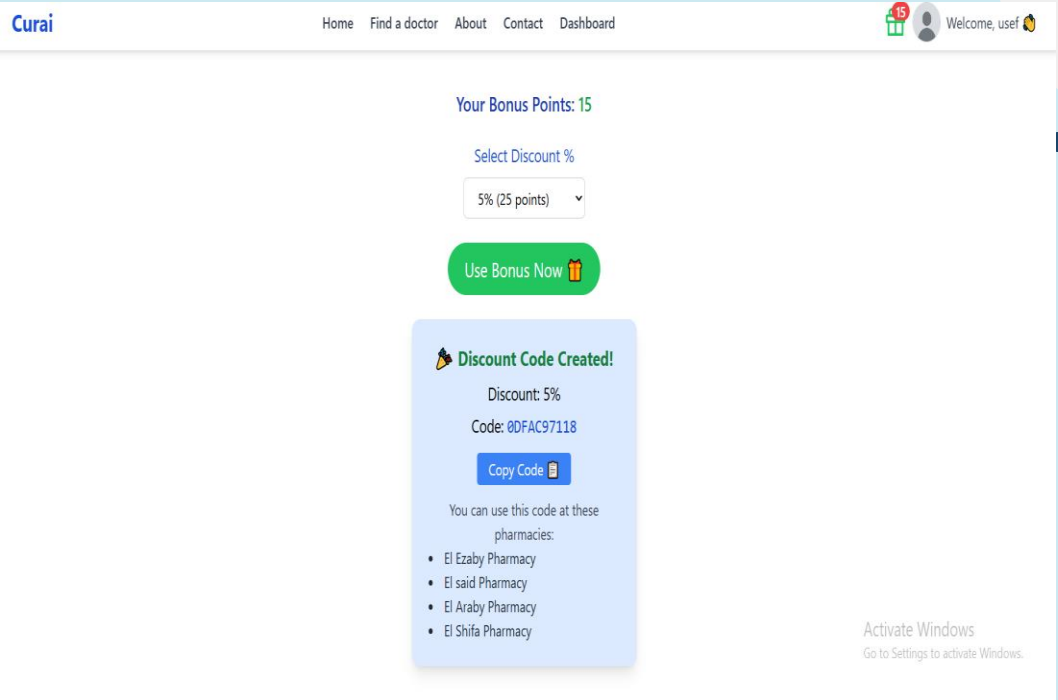
Bonus Page

1. **Discount Code Generation:** The page generates a unique discount code based on the number of points a user has accumulated.

2. **Usage Locations:** It provides a list of places or stores where the discount code can be applied, giving users clear guidance on where they can use their reward.

3. **Reward System:** Designed to enhance user engagement by rewarding points and offering exclusive discounts.

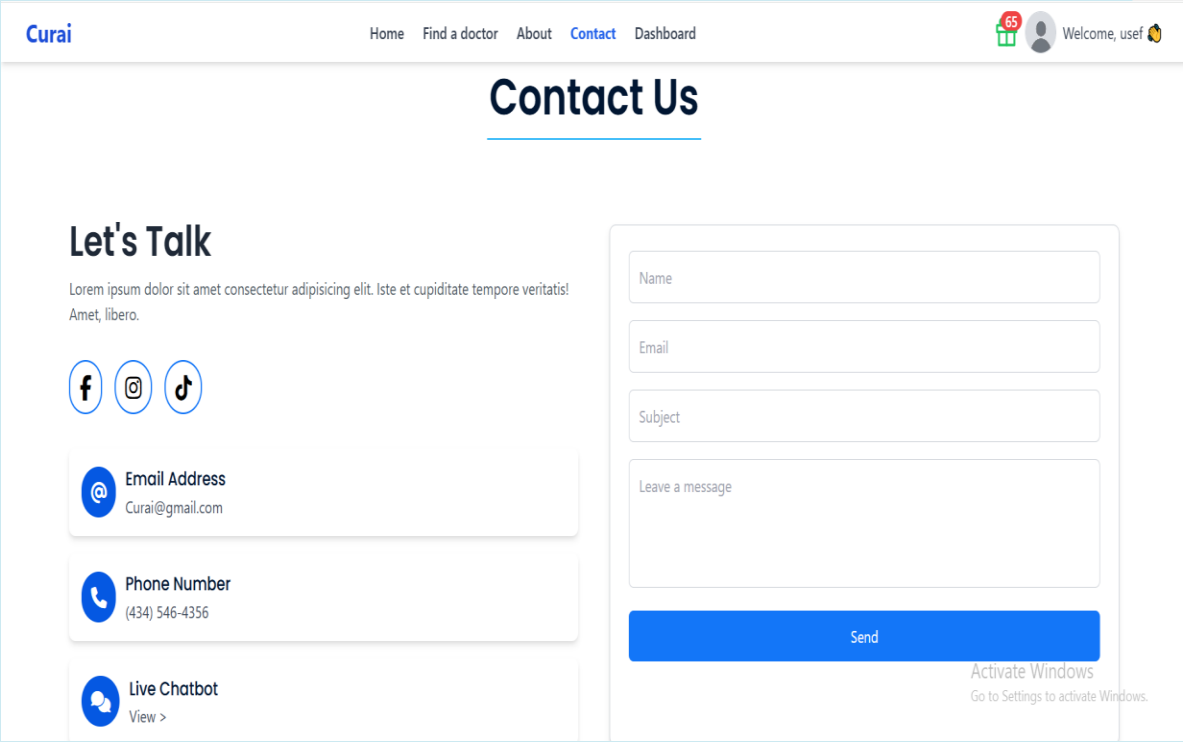
4. **Easy Navigation:** Simple and intuitive interface to allow users to easily view and redeem their rewards.



Contact Page

Contact Page

1. **User-friendly Contact Form:** Allows users to enter their name, subject, email, and message for easy communication.
2. **Multiple Contact Methods:** Includes additional ways to reach out to us, ensuring flexibility in communication.
3. **Social Media Links:** Provides direct links to our active social media accounts for users who prefer to connect through these platforms.
4. **Quick & Convenient:** Designed to ensure users can get in touch with us efficiently, whether via form or social media.



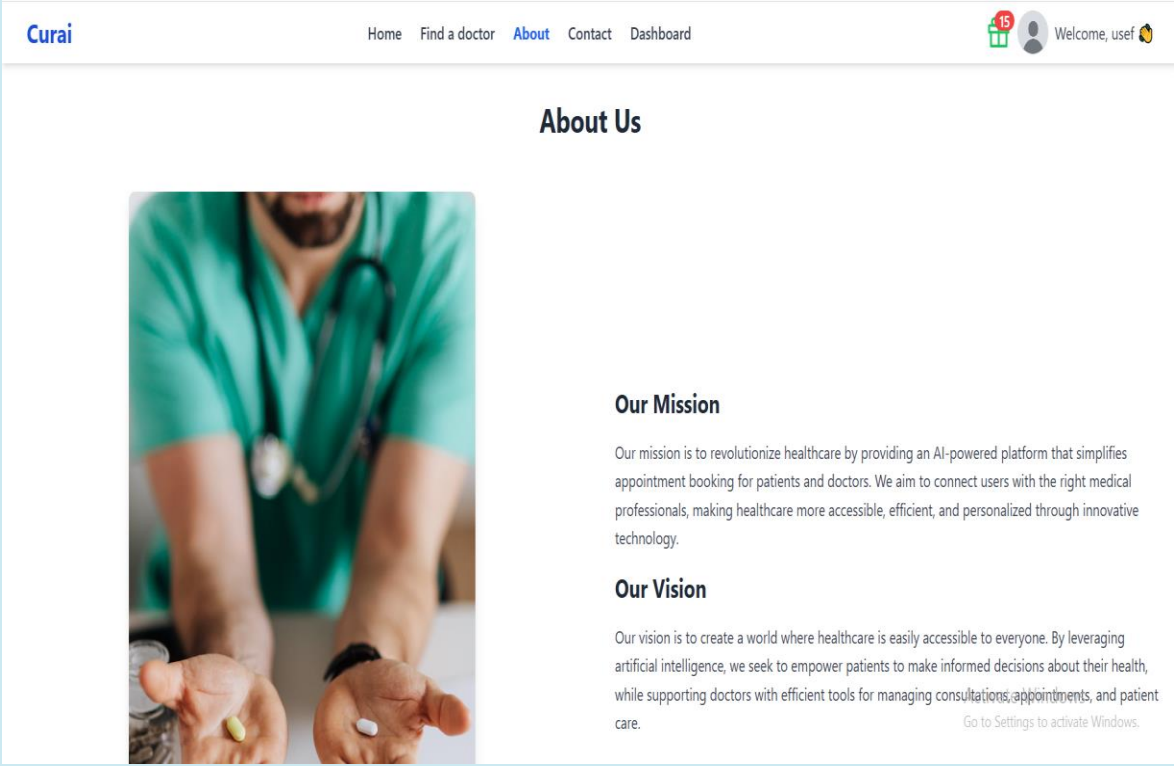
About us Page

1. **The About Page introduces**

2. **Curai**, showcasing our mission to make healthcare more accessible through AI-powered appointment booking. It highlights our commitment to improving the healthcare experience for both patients and doctors by offering innovative and efficient solutions.

3. **Mission:** Making healthcare accessible and personalized through AI.

4. **Vision:** Creating a world with efficient and accessible healthcare for everyone.



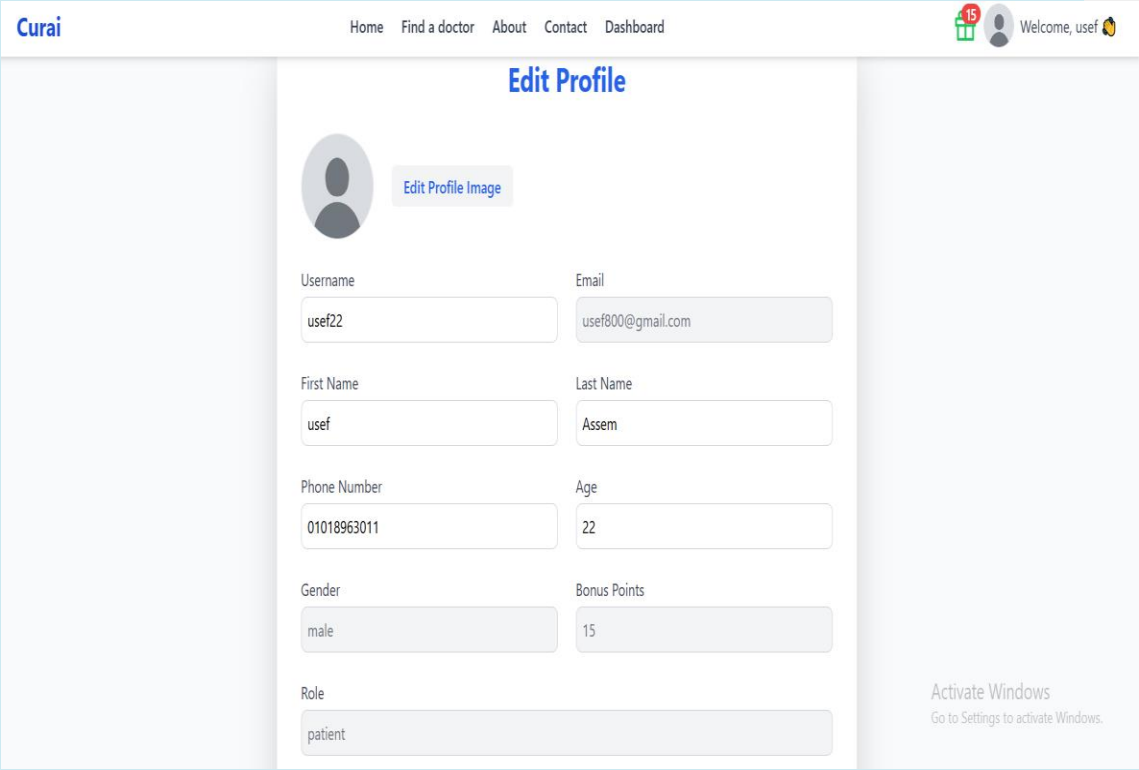
Profile Page

1. **Displays User Information:** Shows **username**, **first name**, **last name**, **email**, **phone number**, **age**, **gender**, **role**, **user image** and **bonus points**.

2. **Editable Fields:** Allows users to update **first name**, **last name**, **phone number**, **user Image** and **age**.

3. **Non-editable Fields: Email**, **gender**, **role**, and **bonus points** cannot be changed.

4. **User-Friendly Interface:** Provides a simple way to manage personal data while ensuring certain details remain secure.



12.Future Work

Enhanced Predictive Models

- Refine AI algorithms to cover a wider range of diseases and health conditions.
- Integrate more comprehensive datasets.
- Apply advanced machine learning techniques to improve prediction accuracy and reliability.

Personalized Medicine

- Develop personalized healthcare plans based on:
- Genetic data.
- Lifestyle factors.
- Health history.
- Provide customized health recommendations and interventions.

UX Enhancement

1. Improve the app’s UI/UX design to be:
2. More intuitive and user-friendly.
3. Accessible to a diverse range of users

Interactive Dashboard

1. Develop a user-friendly dashboard that:
2. Visualizes the user’s health status.
3. Uses rating bars and graphs to track health trends over time.

Data Privacy & Security

- Implement advanced encryption methods.
- Ensure secure data storage.
- Comply with healthcare data protection regulations and standards.

13. Conclusion

Conclusion

- In our presentation, we introduced an intelligent medical application that supports users in identifying possible conditions based on symptoms and helps them book appointments with the right specialists.

- Our goal is to provide accessible, trustworthy, and personalized healthcare support through an easy-to-use interface.

- Moving forward, we aim to expand our medical knowledge base, enhance diagnostic capabilities, and continue refining the user experience – always with a focus on innovation and patient wellbeing.

Curai

Thank You